

A Settlement in Motion

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Grande-Synthe's polder was once part of the sea, sitting below high tide, and its water is expected to keep rising. This project creates a place that lives alongside the water rather than pushing it away: a reversible, gradually adaptive housing model built around a simple idea — stay close to the water, rather than push it away.

About 100 people live here. Houses rise and fall quietly with the water, guided along tall posts sized to the flood levels the site is expected to see, so instead of fleeing, people simply go on living alongside them.

Only the few days when the water rises highest does everyone gather at the one fixed high point, where the market becomes a refuge, reached by whatever means the rising water still allows.

Squares and ponds, fed by a wetland that filters the water upstream, are used for sport, open-air concerts, and waterside gardens most of the time, and simply hold the floodwater when it comes. Rain collected from the rooftops is treated separately and kept as drinking water. The greenery around the site links the lake and the farmland, coexisting with the existing shopping center and stadium while protecting this land's ecology.

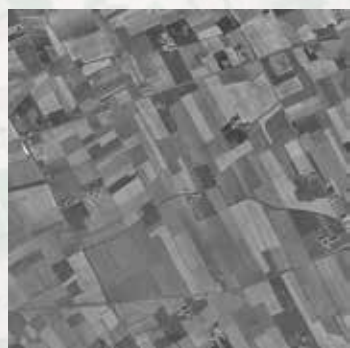
The existing shopping mall is kept, not demolished, woven into the system through shared paths, everyday commerce, rooftop rainwater capture, and a second point of refuge in an emergency. The whole place is built gradually, over roughly ten years, tested and adjusted along the way. Only the refuge and the mall stay fixed, not by default but because keeping people safe and respecting what is already there are reasons worth building around — and it is the people who live here who watch the water and take care of the place, day to day.





The site is located within a stretch of the polder (reclaimed land below sea level) served by the Wateringues, a centuries-old network of artificial drainage canals in northern France. Grande-Synthe lies in the area between this drainage network and the coastline, where the dominant flood risk is coastal flooding from storm surge (submersion marine), rather than river flooding.

Lac du Puythouck was originally a quarry site for extracting sand used in construction materials. After quarrying ended, water accumulated in the excavated pits to form the lake, and natural vegetation gradually returned to the surrounding area—leading to its official designation as a regional nature reserve in 2015. This proposal's site is positioned as a continuation of that same trajectory: a transition from industrial land back to nature.



Excavation
(1950~1965s)



Lake formation
(1965~1980s)



Nature reserve
(Present)

A Mixed-Use Residential District

— Realized through five evaluation criteria that bring housing and public space together as one integrated structure.

Zero extra flood risk

The Wateringues' drainage network has little margin left, and record rainfall keeps pushing it closer to its limit. Storing rainwater on-site lets the site expand housing without adding new load to that system.

Hold water, longer

Water plazas, ponds, and cisterns are distributed across the site and linked in a cascade, filling and overflowing in sequence. This uses the full storage capacity without waste, easing flood risk for the surrounding area.

Reconnected ecology

Green belts along the woodland and road edges create connection points with the existing nature reserve network. A fully continuous corridor remains a goal for future design development.

Keep the ground permeable

Elevated housing and permeable pond beds leave the ground surface untouched, and roof rainwater feeds directly into the site's water system. Most of the site stays as exposed soil, letting rainwater return naturally into the ground.

One water system, both sides

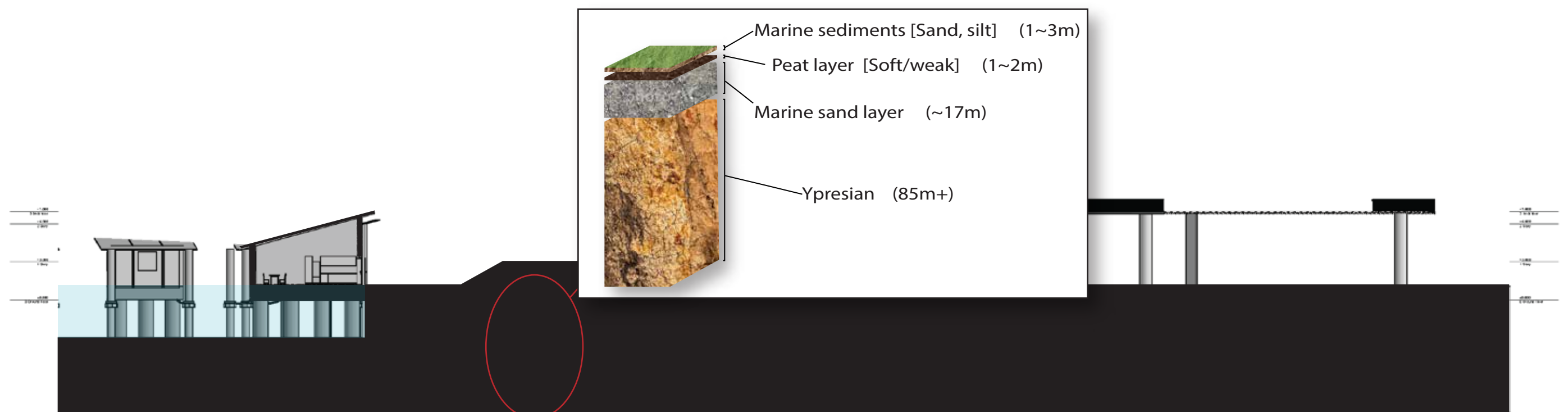
On land, the water plaza serves both daily use and flood storage; on the lake, boats link the site to Base nautique for everyday transport and emergency evacuation. The lake becomes something both to look at and to use.



The flooding scenario shows the water-depth distribution around the site, based on data from Géorisques—the French government's official risk assessment database (TRI: Territoire à Risque important d'Inondation, 2020). According to current official data, the center of Grande-Synthe is not directly included within the designated storm-surge flooding zones, but it does fall within the boundaries of a TRI (an area identified as being at significant flood risk). As sea-level rise and the intensification of storm surges progress, these projections may be updated in the future, and this proposal is positioned as a design that anticipates and responds to that potential future expansion of risk.



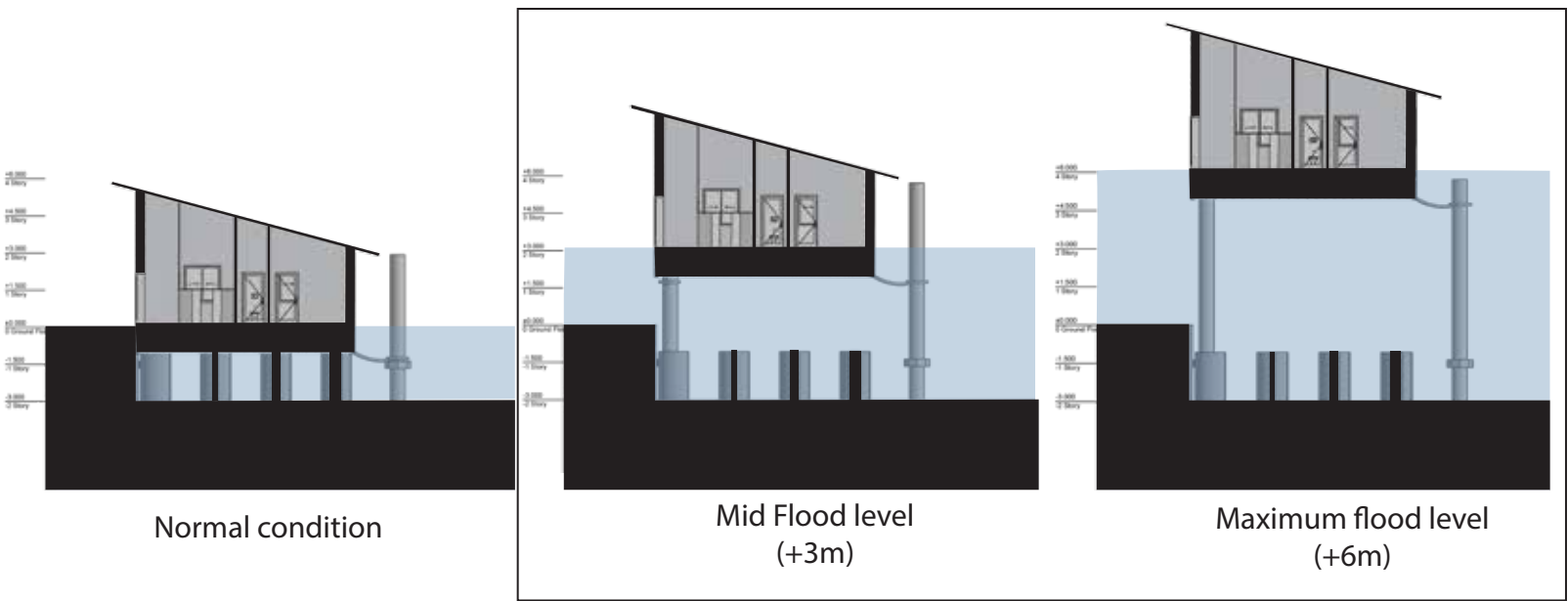
Lac du Puythouck, adjacent to the site, is part of the Réserve naturelle régionale de Grande-Synthe (Grande-Synthe Regional Nature Reserve), officially designated by the Hauts-de-France Regional Council in July 2015, and constitutes the largest sector within the reserve as a whole (approximately 174 hectares). Farmland and woodland extend to the west of the site, while roads and existing commercial facilities spread to the east and north—leaving the reserve and the farmland divided from each other by this urbanized area. The ecological corridor proposed here is a first step intended to reconnect this fragmented ecological network, but at this stage it remains limited to part of the site (the woodland-facing and road-facing edges) and does not yet form a continuous corridor.



The site's design water levels are set at two stages: a frequently occurring, standard-scale flood (+3.0m), and a more severe storm-surge flooding scenario (+6.0m). This two-stage approach reflects the characteristics of storm-surge flooding (submersion marine)—the dominant flood type along France's northern coast. The deck floor height is set at +7.0m, providing more than 1m of safety margin (freeboard) above this projected maximum water level (+6.0m). This is also consistent with the threshold used in France's flood risk zoning system (PPRI), under which areas with water depths exceeding 1m are designated as the highest-risk "red zone."

For the housing's foundation system, a "raised earth platform" — building up the foundation with mounded soil — was initially considered. However, the site consists of soft ground made up of peat layers and marine sediments, and it became clear that this approach carried risks of slope erosion as well as differential settlement caused by the concentration of building loads at a single point. Based on this assessment, the housing form was unified around a stilt-based system that lifts the buildings on piles without mounding the ground (movable piles plus a floating floor), rather than an earthen platform. Each house is composed of vertically fixed movable piles (guide posts) and a floating floor (a hollow concrete hull) that slides up and down around them. The hull's outer shell is built of reinforced concrete, with expanded polystyrene (EPS) filled inside as a reserve buoyancy source—creating a dual safety structure that maintains buoyancy even if the outer shell is ever damaged. Under normal conditions, the houses sit along the edge of the irregularly shaped storage basin (3m deep), forming a unified landscape together with the surrounding hill and floating gardens. As water levels rise, the houses rise along the guide posts, so that even at the projected maximum water level (+6.0m), the living spaces remain above water. When the water level falls, the houses descend back to their original position under their own weight. A few terminology notes: I rendered movable piles and guide posts (kept as a parenthetical gloss, matching your original structure), and floating floor, consistent with the floating-foundation terminology used earlier. Let me know if you'd prefer guide piles instead of movable piles for consistency with any structural engineering terms used elsewhere.

The terp (evacuation hub) is planned as a ring-shaped observation platform (at least 15,000mm wide). The upper level functions as an observation terrace, serving under normal conditions as a space for strolling and lingering with panoramic views of the lake. The lower level provides direct access to the water surface, serving for boat landings as well as a vertical evacuation route in emergencies. Through this two-tier upper/lower structure, the terp simultaneously fulfills both a "viewing" function and a "using" function.



This is a emergen level. when the water overrisd up, the access is changed by boat.





Phase 1 : Infrastructure

Construct the main deck, storage basins, and hill; install the water-level monitoring system.



Because a foundation for capturing water and safe circulation routes must be established before housing begins.



Phase 2 : First Housing Cluster + Terp

Install the first phase of housing and the terp (evacuation hub).



Because evacuation measures must be in place from the moment residents begin occupying the site.



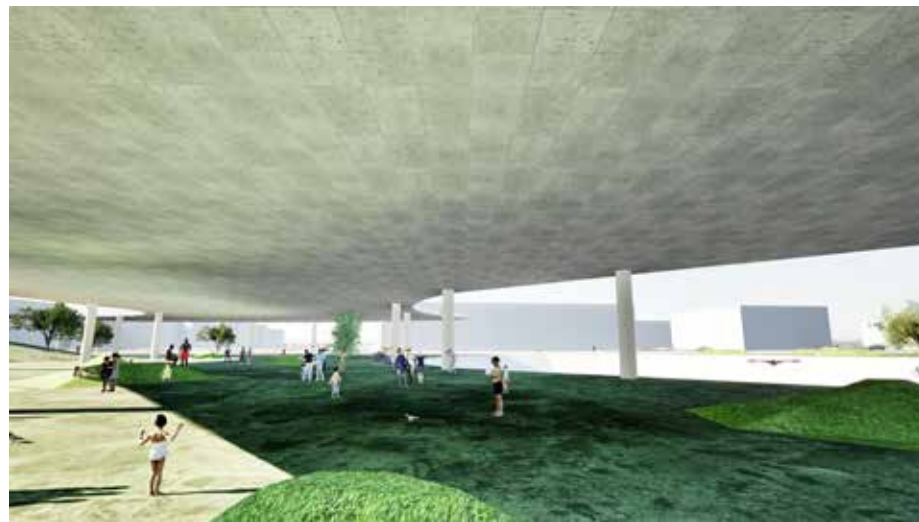
Phase 3 : Full Completion

Complete the remaining housing and finish the ecological corridor.

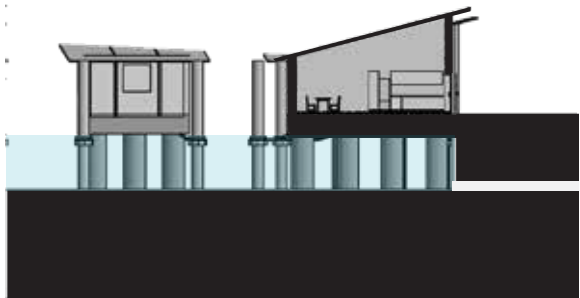
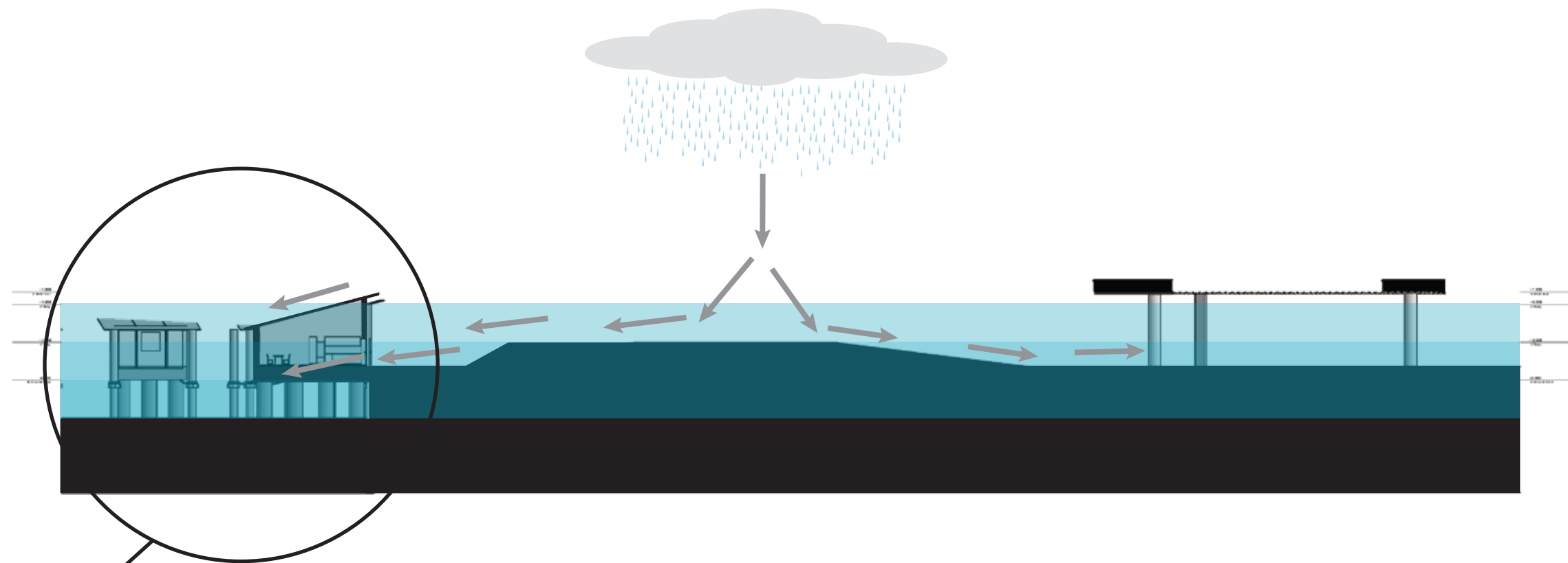


Because ecological considerations must be completed in step with the growing population.

The hill (3m at its summit, built up using excavated soil) functions as a park and botanical garden, as well as a vegetable garden for residents to cultivate. During floods that exceed the design threshold, it also serves as a high-ground refuge that stays above water. The oval-shaped area combines water storage with a floating garden and a track-like walking path, forming a multi-functional space. The irregularly shaped cluster of storage basins (3m deep) provides both the storage that supports the housing's floating foundations and space for the floating gardens; the basins themselves are not primarily intended for everyday human access. Multiple housing units are arranged along the edges of these basins, each accessed individually via a movable pile and gangway. The space along the side of the deck (15m wide) includes fixed temporary kiosks and gently raised mounds (50–100cm high) where people can lie down and relax. In daily use, it serves as a place for rest, dining, and viewing; during events, it becomes a lively gathering space. These functions are allowed to be temporarily lost during large-scale flooding, with permanent disaster-resilience functions instead carried by the terp and the housing's floating structures. A few terminology notes: I kept "terp" as-is (consistent with earlier usage), and rendered as "floating foundations and structures," which is standard English terminology for buoyant/amphibious foundation systems. Let me know if any of these technical terms should align with specific engineering vocabulary you're using elsewhere in the document. The projected residential population is approximately 100 people (scaled down from the initial plan of 280, in consideration of the load burden on the floating foundations). In addition to this residential population, the deck is designed as a public space intended to accommodate up to 500 or more users, including visitors.



The site handles two types of water: rainwater and floodwater that has entered the site. Rainwater falling on the roofs passes through green, purifying roof layers (planted with native species such as willow and reed), where it is purified before being stored and reused. Water that enters buildings during flooding is pumped up to these same roof planting layers, purified in the same way, and then either discharged or returned to storage.



Water on the site is first received by the irregularly shaped cluster of storage basins (3m deep, functioning as a flood-control buffer). Rather than being connected by visible channels, these basins are linked to one another through submerged connecting pipes, allowing water to move naturally between them driven by differences in water level (hydrostatic pressure)—a power-free "cascade" mechanism. The surrounding trees and wetlands supplement this buffering function through infiltration, evapotranspiration, and water purification. This stored water is inherently "temporary" in character: over time, it is gradually released beyond the site and into the lake through this natural, level-driven cascade. Ultimately, some of the purified water is reused on-site, while the remainder flows gradually out of the site.



Aushopping Grande Synthe

Observation Deck

Housing Cluster

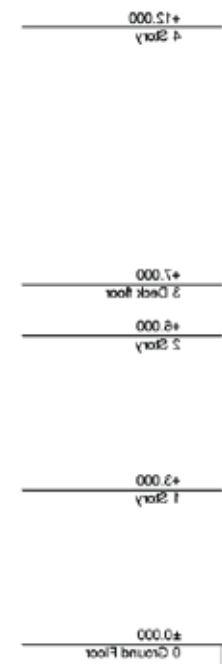
Water Basins

The Puythouck Lake

Excavated Mound

Ecological Corridor

Forest

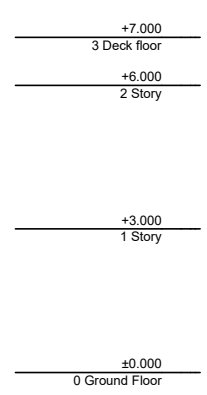


Acetylated timber (30mm)

Unlike conventional preservative-treated timber, acetylated wood is chemically modified at the molecular level, achieving high water resistance and durability without relying on chemical preservatives. This makes it well suited to a site where water levels fluctuate repeatedly, reducing long-term maintenance needs.

Reinforced concrete [RC] (200mm)

Order	Material	Thickness
1	Willow (Salix) + Reed (Phragmites)	—
2	Soil substrate	250 mm
3	Reinforced concrete (RC)	200 mm



Order	Material	Thickness
1	Fibre-cement siding	15 mm
2	Air cavity (furring strip)	30 mm
3	Rubber/asphalt-based sheet	2 mm
4	Extruded polystyrene (XPS)	100 mm
5	Reinforced concrete (RC)	180 mm
6	Gypsum board	12.5 mm
7	timber finishing	2 mm

Order	Material	Thickness
1	timber finishing	2 mm
2	Gypsum board	12.5 mm
3	Light Gauge Steel (LGS), 65 type	65 mm
4	Glass wool (substituted with Air material in model)	50–65 mm
5	Gypsum board	12.5 mm
6	timber finishing	2 mm

LGS (Light Gauge Steel)

► Partition walls are non-load-bearing, so ease of construction and light weight are prioritized over structural strength. LGS offers high workability and adapts easily to future interior layout changes.

EPS (Expanded Polystyrene)

► Provides a backup source of buoyancy inside the hull, ensuring the house remains afloat even if the primary RC shell is damaged — a dual-safety approach to the floating foundation.

XPS (Extruded Polystyrene)

► Rigid foam insulation with very low water absorption, since its cells are fully fused during extrusion. Used at the exterior wall and floor, where water contact is most likely — distinct from EPS (hull buoyancy) and glass wool (interior partitions, away from water).

Reinforced concrete [RC] (800mm)

Epoxy resin coating (2.0mm)

Order	Material	Thickness
1	Timber (rot-resistant treatment)	20 mm
2	Structural plywood	24 mm
3	Rubber/asphalt-based sheet	2 mm
4	Extruded polystyrene (XPS)	100 mm
5	Reinforced concrete (RC)	150 mm
6	Air cavity	≈ 160 mm
7	Expanded Polystyrene (EPS)	≈ 600 mm
8	Reinforced concrete (RC)	150 mm

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